2nd REVIEW REPORT

Date : 17/08/2026

Response to the remarks of the 1st review:

As recommended in the first review, we transitioned from preliminary conceptual models toward a dedicated mobile architecture with a structured behavioral data pipeline. We addressed feedback regarding habit estimation by planning a transition away from initial hard-coded if-else logic toward formal feature extraction and machine learning classification.

Work done till the 1st review:

Completed the comprehensive literature survey on mHealth smoking cessation tools and defined system requirements for edge-based gesture detection. Designed the core application UI wireframes in Flutter, created the initial Supabase database schema, and established baseline manual puff logging. Implemented initial hard-coded if-else logic based on historical puff counts to verify basic tracking functionality.

Work done after the 1st review:

Set up the cross-platform Flutter project architecture, implemented Supabase authentication, and created data models for puff events and sessions. Formulated dataset collection strategies for hand-to-mouth vape and cigarette gestures for YOLOv9 training. Designed the initial pipeline to log temporal behavioral metadata, such as inter-puff intervals and daily session clustering.

Work to be done before 3rd review:

Train and quantize the YOLOv9-tiny object detection model for on-device inference using TensorFlow Lite. Replace heuristic rules with a Decision Tree model for addiction risk classification and linear regression for craving prediction. Integrate live camera feeds via Dart Isolates and develop comprehensive dashboard visualization and craving relief modules.

Remarks of the Guide:

Architectural design and database schemas are well structured. Proceed with dataset collection and machine learning pipeline implementation as planned.

Probable Date of Completion of Project Work: 25/11/2026

Signature of the students	Signature of the Guide	Signature of Project Coordinator

Review Remarks:

Good progress on mobile framework setup and database schema; ensure ML pipeline replaces rule-based heuristics before next review.

Name & Signature of Review Committee Members		

Weekly Report (24/08/2026 to 28/08/2026)

Date : 28/08/2026

Work done in the Previous Week:
Completed approximately 40% of the behavioral metadata extraction and addiction classification pipeline. Conducted initial work on the Decision Tree based severity classification by defining key feature metrics, including daily puff frequency and inter-puff intervals. Prepared and structured the annotated video dataset to train the computer vision gesture recognition model.
Remarks of the Guide:
Good progress on behavioral feature extraction. Ensure the feature thresholds for the Decision Tree align with clinical addiction severity criteria.

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Weekly Report (31/08/2026 to 04/09/2026)

Date : 04/09/2026

Work done in the Previous Week:
Trained the YOLOv9-tiny model on hand-to-mouth gestures and exported the quantized TFLite weights for mobile deployment. Integrated the TFLite inference engine into Flutter using background Dart Isolates to maintain a smooth 60 FPS camera preview. Implemented spatial mouth proximity boundaries and a 1.2-second dwell state machine to filter false-positive detections.
Remarks of the Guide:
Real-time inference integration using Isolates is commendable. Test edge detection under diverse ambient lighting conditions.

Signature of the students	Signature of the Guide	Signature of Project Coordinator

Weekly Report (07/09/26 to 25/09/26)

Date : 25/09/2026

Work done in the Previous Week:

Finalized the Decision Tree addiction classifier to categorize users into Low, Moderate, and High severity tiers based on multi-day usage density. Developed the Multivariate Linear Regression model to forecast upcoming craving windows from historical session intervals. Integrated local background push notifications to proactively alert users before forecasted craving peaks.

Remarks of the Guide:

The integration of the predictive craving engine with local notifications is satisfactory. Proceed to user interface and analytics dashboard integration.

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Weekly Report (05/10/26 to 09/10/26)

Date : 09/10/2026

Work done in the Previous Week:

Implemented interactive data visualization charts using fl_chart to display daily/weekly puff frequencies and session intervals. Developed craving relief utilities, including an animated 4-7-8 diaphragmatic breathing timer and behavioral urge distraction techniques. Conducted unit and widget testing across Supabase sync, video analysis, and ML feature calculation modules.

Remarks of the Guide:

Dashboard visualization and craving relief tools are nicely built. Prepare the consolidated system demonstration for the upcoming 3rd review.

Signature of the students	Signature of the Guide	Signature of Project Coordinator

3rd REVIEW REPORT

Date : 31/10/26

Response to the remarks of the 2nd review:
Fully eliminated legacy hard-coded heuristic logic by deploying trained ML models for addiction classification and craving forecasting. Successfully optimized on-device inference speed by isolating camera frame tensor preprocessing, maintaining 60 FPS UI rendering without dropped frames.
Work done till the 2nd review:
Completed system requirements, UI design, Supabase database integration, and foundational puff tracking workflows. Prepared the custom gesture dataset and framed the algorithmic structure for behavioral metadata extraction.
Work done after the 2nd review:
Trained, converted, and integrated the YOLOv9-tiny TFLite model with spatial and temporal dwell-time filtering for real-time puff detection. Implemented the Decision Tree severity classifier and linear regression craving predictor operating entirely on client-side behavioral metadata. Built comprehensive analytics dashboards, automated push alerts, and animated 4-7-8 breathing relief tools.
Work to be done:
Perform comprehensive field testing and latency benchmarking across different Android hardware specifications. Gather qualitative usability feedback on craving intervention effectiveness and finalize project documentation and the final report.
Remarks of the Guide:
All primary modules—edge vision detection, ML analytics, and user intervention—are functional and well integrated. Approved to proceed toward final report submission.

Probable Date of Completion of Project Work: 25/11/2026

Signature of the students	Signature of the Guide	Signature of Project Coordinator

Review Remarks:
End-to-end working system demonstrated successfully. Proceed with final user evaluation and complete thesis documentation.

Name & Signature of Review Committee Members		